

Eukaryotes, Prokaryotes & the Scale of Cells

Test · 24 marks total

Name: _____

Date: _____

Score: _____ / 24

Section A — Eukaryotic or prokaryotic?

State whether each description is eukaryotic or prokaryotic.

- (a) A cell with genetic material enclosed in a nucleus.
- (b) A cell with genetic material as a single DNA loop, not enclosed in a nucleus.
- (c) A cell with plasmids.
- (d) A cell with mitochondria.

[4 marks]

Section B — Bacterial cell structure

B1. List the three structures found in a bacterial cell.

[3 marks]

B2. State what a plasmid is.

[1 mark]

Section C — Unit conversion

C1. Convert 0.05 mm to micrometres.

[2 marks]

C2. Convert 3000 nanometres to micrometres.

[2 marks]

C3. Convert 2 micrometres to millimetres.

[2 marks]

Section D — Standard form

D1. A bacterium is 2 μm across. Write this in metres, using standard form.

[2 marks]

D2. An animal cell is 25 μm across. Write this in metres, using standard form.

[2 marks]

D3. Convert 4×10^{-5} m to micrometres.

[2 marks]

Section E — Order of magnitude

E1. A bacterium is about $2\text{ }\mu\text{m}$ across and an animal cell is about $20\text{ }\mu\text{m}$ across. Calculate how many times bigger the animal cell is, and state this as an order of magnitude.

[2 marks]

E2. A plant cell is $4 \times 10^{-5}\text{ m}$ wide and a bacterium is $2 \times 10^{-6}\text{ m}$ wide. Calculate how many times wider the plant cell is.

[2 marks]